



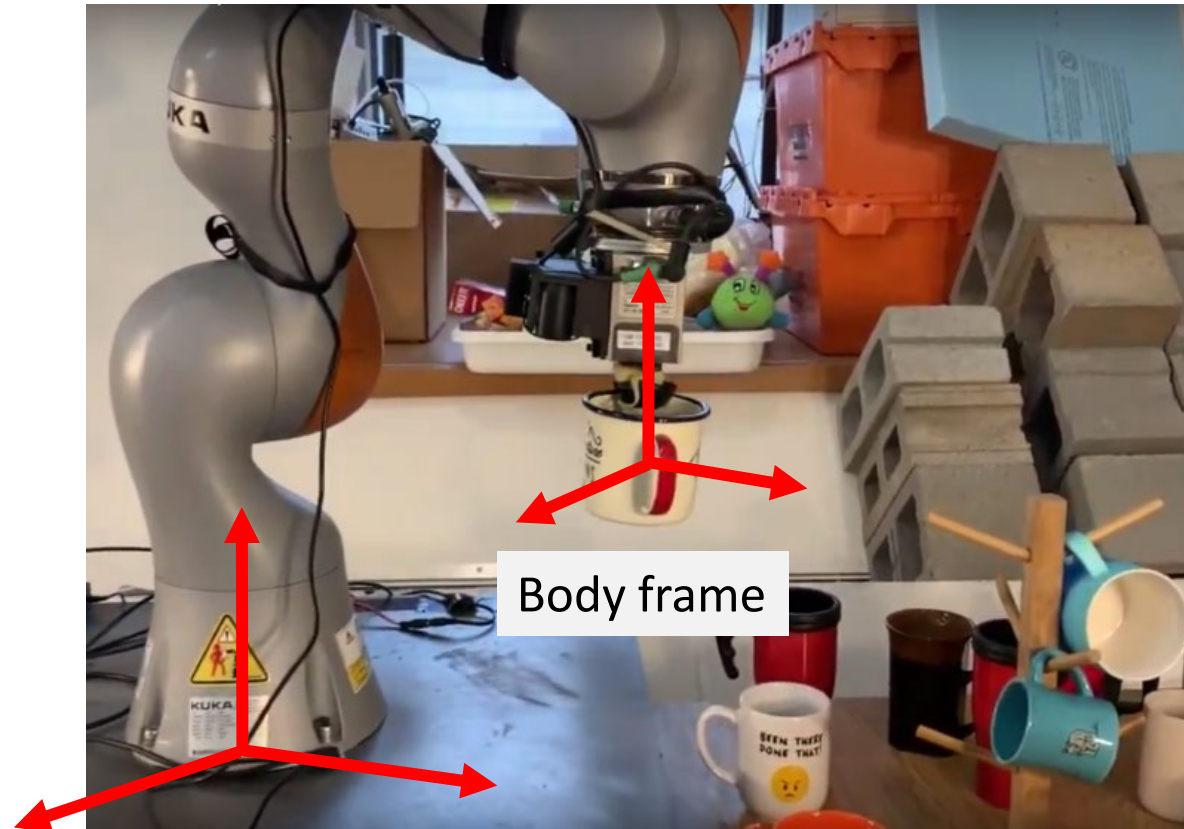
Rigid-Body Motions and Transformations

CS 6341 Robotics

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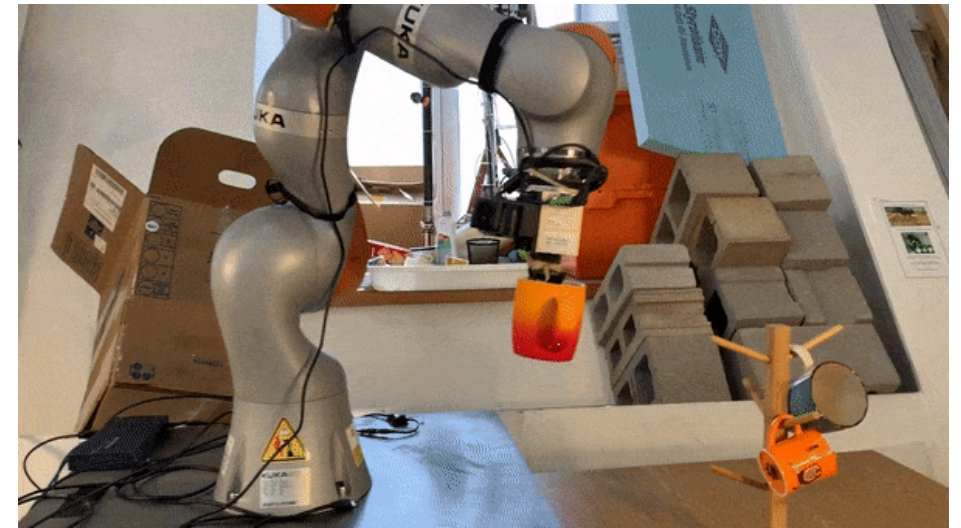
The University of Texas at Dallas

Rigid-Body Motions



Body frame

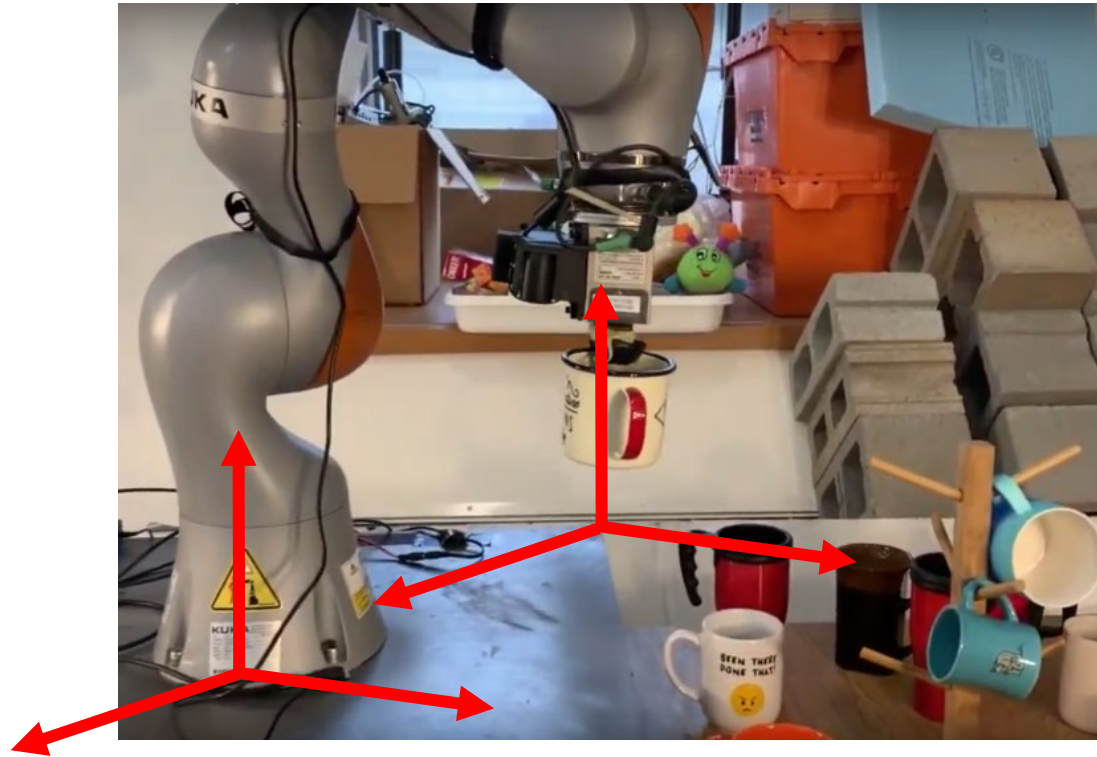
Space frame (fixed frame)



<https://venturebeat.com/ai/mit-csail-refines-picker-robots-ability-to-handle-new-objects/>

Reference Frames

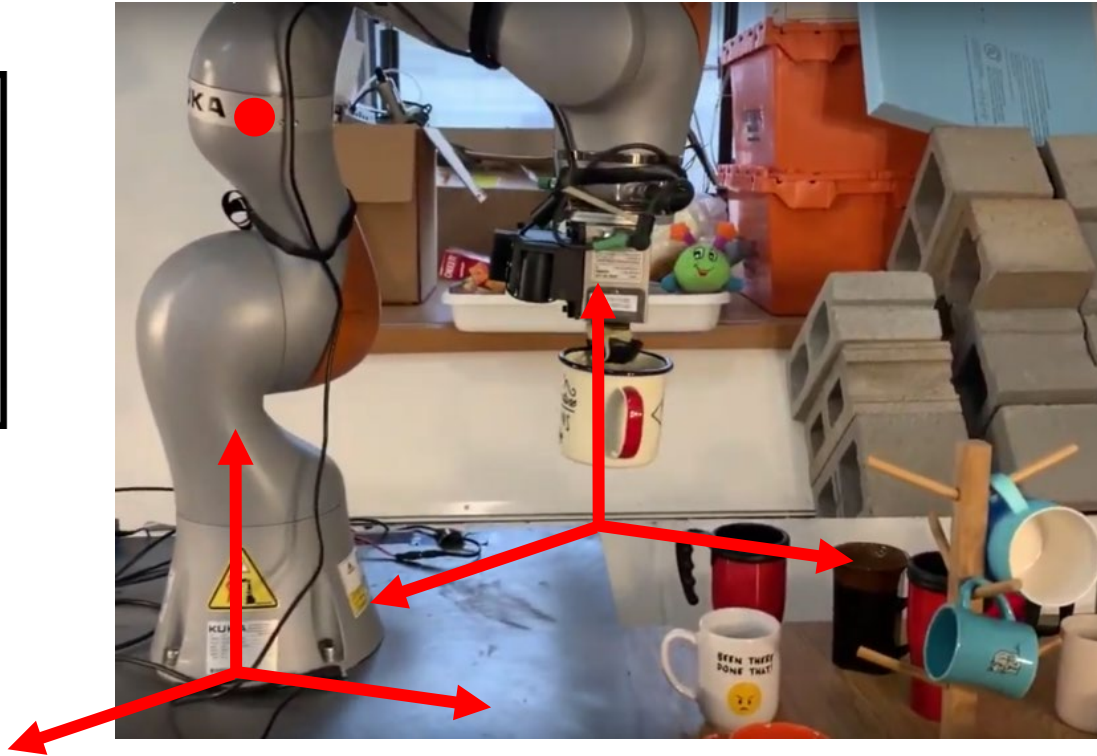
- A reference frame can be attached anywhere



More about Reference Frames

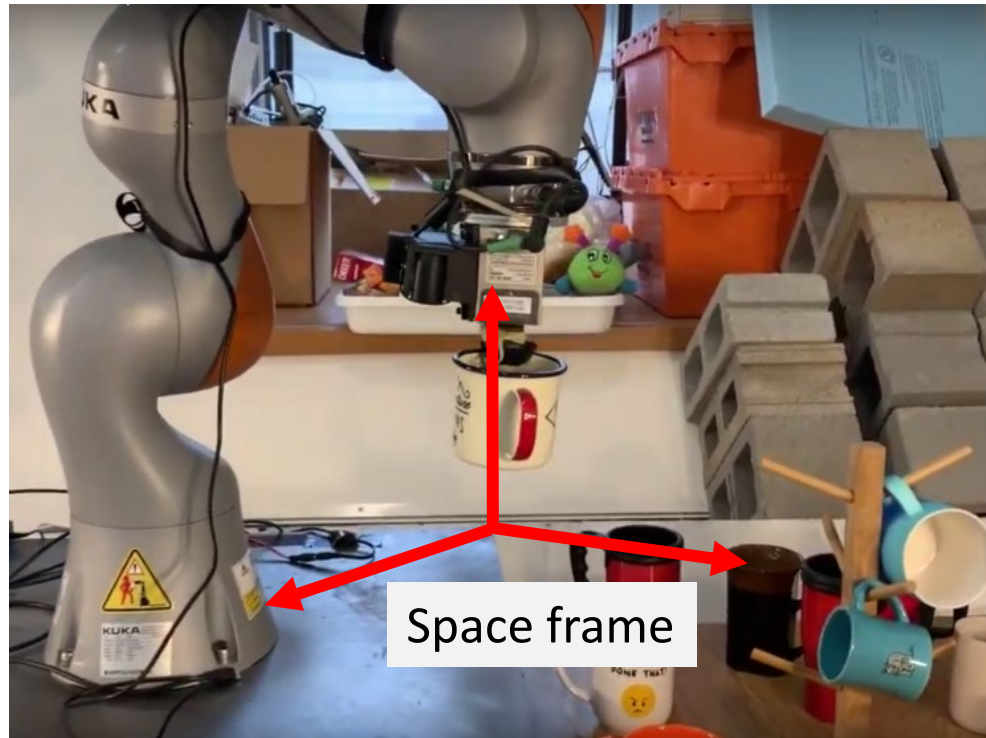
- Different reference frames result in different representations of the space and objects, but the underlying geometry is the same

$$\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$



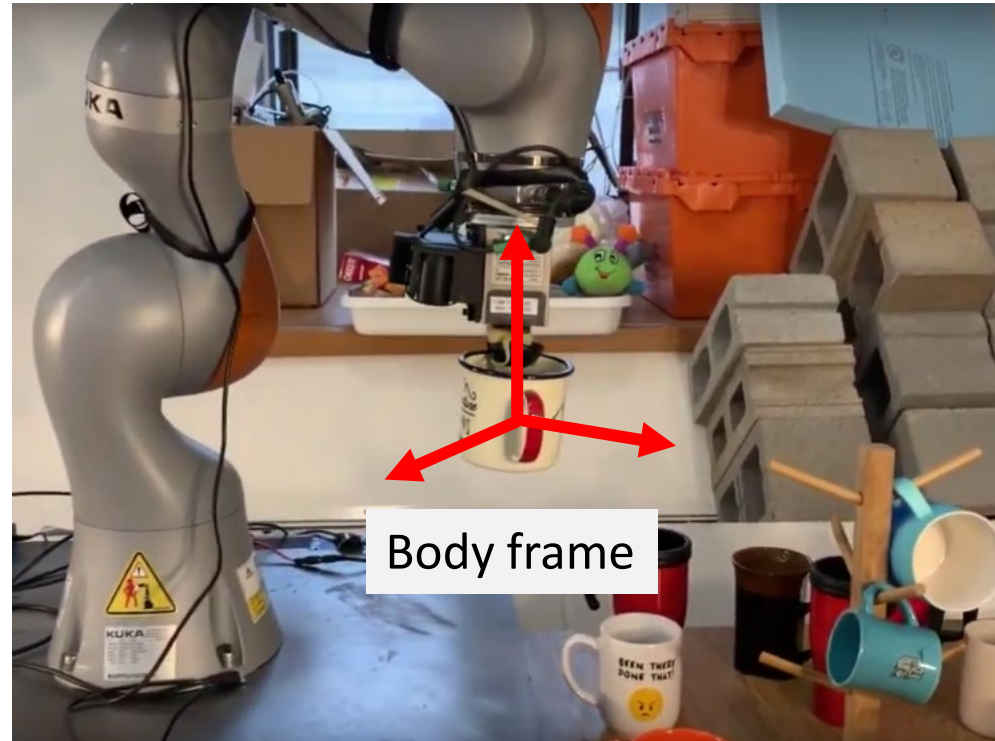
More about Reference Frames

- Always assume one stationary **fixed frame** or **space frame** $\{s\}$
 - E.g., a corner of a room



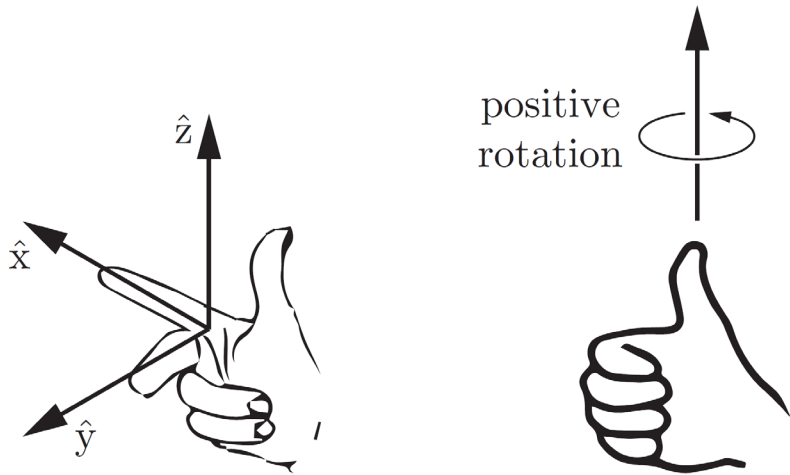
More about Reference Frames

- **Body frame** {b} has been attached to some moving rigid body
 - E.g., origin on the center of mass of the body
 - No need to be on the physical body!



More about Reference Frames

- All frames in this course are stationary, inertial frames
 - Body frame is a **motionless** frame that is instantaneously coincident with a frame that is fixed to (possibly moving) body
- All frames in this course are right-handed



- Point your **right thumb along the positive direction of the rotation axis**.
- Curl your other four fingers.
- The direction your fingers curl is the **positive direction of rotation**.

Rigid-Body in 3D

$$p = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}$$

- Origin of the body frame

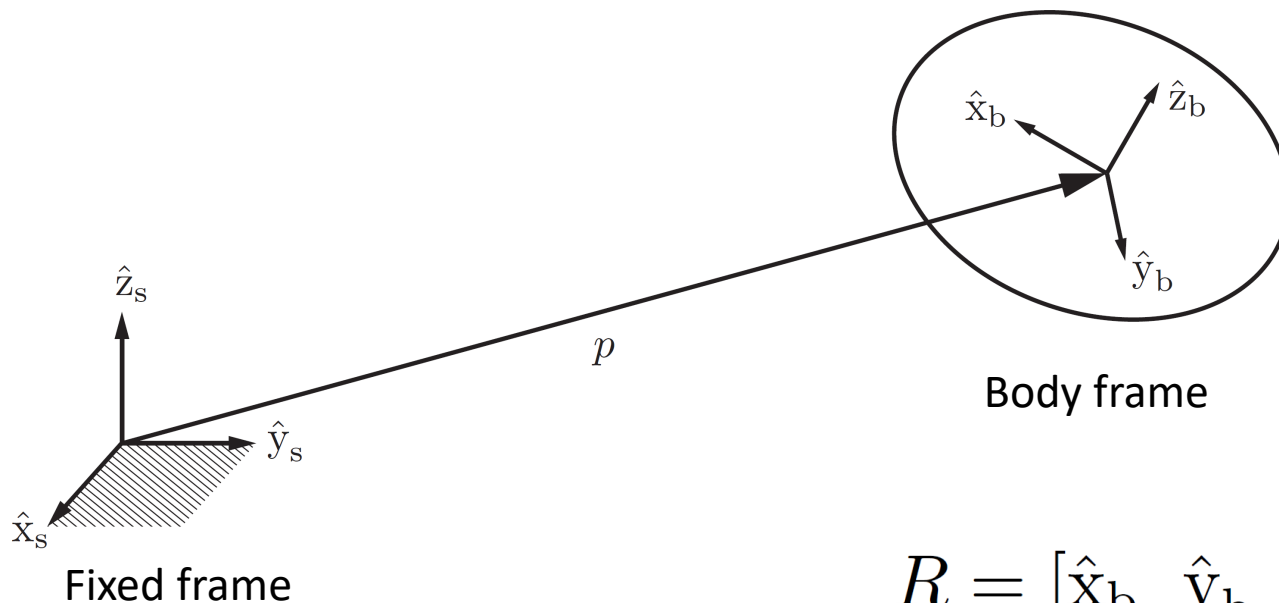
$$p = p_1 \hat{x}_s + p_2 \hat{y}_s + p_3 \hat{z}_s$$

- Axes of the body frame

$$\hat{x}_b = r_{11}\hat{x}_s + r_{21}\hat{y}_s + r_{31}\hat{z}_s,$$

$$\hat{y}_b = r_{12}\hat{x}_s + r_{22}\hat{y}_s + r_{32}\hat{z}_s,$$

$$\hat{z}_b = r_{13}\hat{x}_s + r_{23}\hat{y}_s + r_{33}\hat{z}_s.$$



$$R = [\hat{x}_b \ \hat{y}_b \ \hat{z}_b] = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \quad \text{Write as column vectors}$$

Rotation matrix

Rotation Matrix

- Unit norm condition

$$r_{11}^2 + r_{21}^2 + r_{31}^2 = 1,$$

$$r_{12}^2 + r_{22}^2 + r_{32}^2 = 1,$$

$$r_{13}^2 + r_{23}^2 + r_{33}^2 = 1.$$

$$R = [\hat{x}_b \quad \hat{y}_b \quad \hat{z}_b] = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

- Orthogonality condition $\hat{x}_b \cdot \hat{y}_b = \hat{x}_b \cdot \hat{z}_b = \hat{y}_b \cdot \hat{z}_b = 0$

$$r_{11}r_{12} + r_{21}r_{22} + r_{31}r_{32} = 0,$$

$$r_{12}r_{13} + r_{22}r_{23} + r_{32}r_{33} = 0,$$

$$r_{11}r_{13} + r_{21}r_{23} + r_{31}r_{33} = 0.$$

Properties of Rotation Matrices

- Closure $R_1 R_2$
- Associativity $(R_1 R_2) R_3 = R_1 (R_2 R_3)$
- Identity element: identity matrix I
- Inverse element $R^{-1} = R^T$
- Not commutative $R_1 R_2 \neq R_2 R_1$

Rotation Representations

- A 3x3 matrix

$$R = [\hat{x}_b \quad \hat{y}_b \quad \hat{z}_b] = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

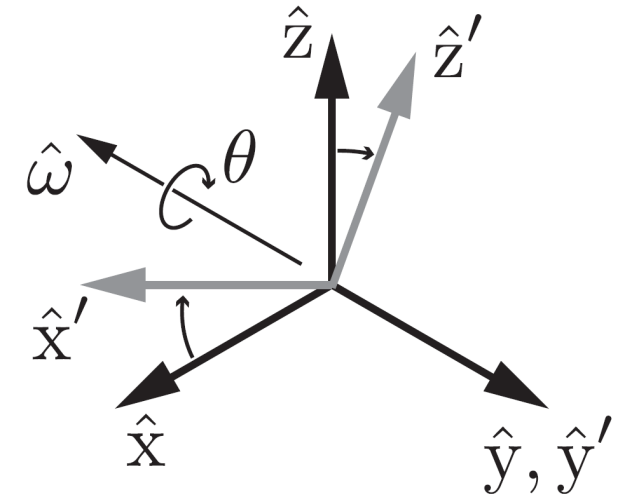
- Axis-angle $R = \text{Rot}(\hat{\omega}, \theta)$ $\hat{\omega} = (\hat{\omega}_1, \hat{\omega}_2, \hat{\omega}_3)$

$$\text{Rot}(\hat{\omega}, \theta) =$$

$$\begin{bmatrix} c_\theta + \hat{\omega}_1^2(1 - c_\theta) & \hat{\omega}_1\hat{\omega}_2(1 - c_\theta) - \hat{\omega}_3s_\theta & \hat{\omega}_1\hat{\omega}_3(1 - c_\theta) + \hat{\omega}_2s_\theta \\ \hat{\omega}_1\hat{\omega}_2(1 - c_\theta) + \hat{\omega}_3s_\theta & c_\theta + \hat{\omega}_2^2(1 - c_\theta) & \hat{\omega}_2\hat{\omega}_3(1 - c_\theta) - \hat{\omega}_1s_\theta \\ \hat{\omega}_1\hat{\omega}_3(1 - c_\theta) - \hat{\omega}_2s_\theta & \hat{\omega}_2\hat{\omega}_3(1 - c_\theta) + \hat{\omega}_1s_\theta & c_\theta + \hat{\omega}_3^2(1 - c_\theta) \end{bmatrix}$$

$$s_\theta = \sin \theta \quad c_\theta = \cos \theta$$

Rodrigues' formula



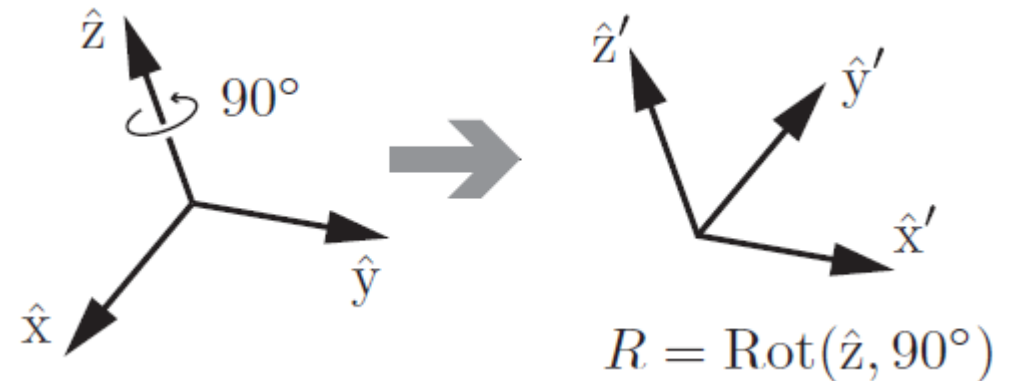
Rotation Representations

- Axis-angle

$$\text{Rot}(\hat{x}, \theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

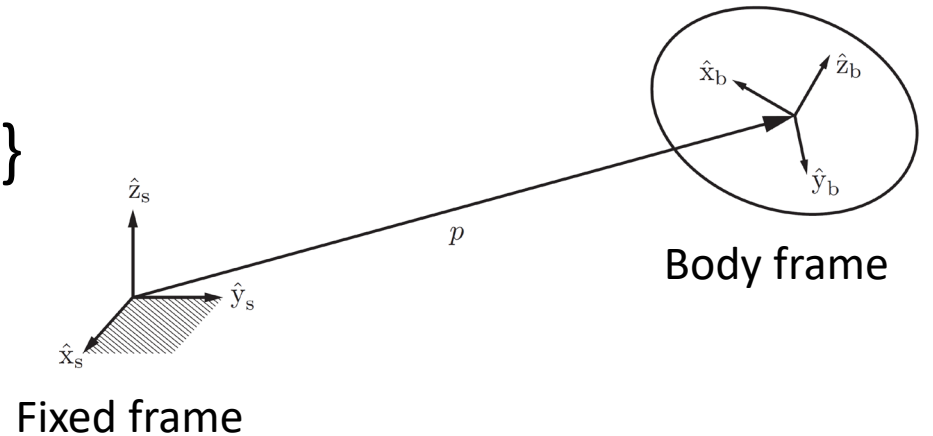
$$\text{Rot}(\hat{y}, \theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$\text{Rot}(\hat{z}, \theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Homogeneous Transformation Matrices

- Consider body frame {b} in a fixed frame {s}
 - 3D rotation $R \in SO(3)$
 - 3D position $p \in \mathbb{R}^3$



- Special Euclidean group SE(3) or homogenous transformation matrices

$$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_1 \\ r_{21} & r_{22} & r_{23} & p_2 \\ r_{31} & r_{32} & r_{33} & p_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Properties of Transformation Matrices

- The inverse of a transformation matrix is also a transformation matrix

$$T^{-1} = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix}$$

- Closure $T_1 T_2$
- Associativity $(T_1 T_2) T_3 = T_1 (T_2 T_3)$
- Identity element: identity matrix I
- Not commutative $T_1 T_2 \neq T_2 T_1$

Homogeneous Coordinates

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

homogeneous image
coordinates

$$(x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = w \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

homogeneous scene
coordinates

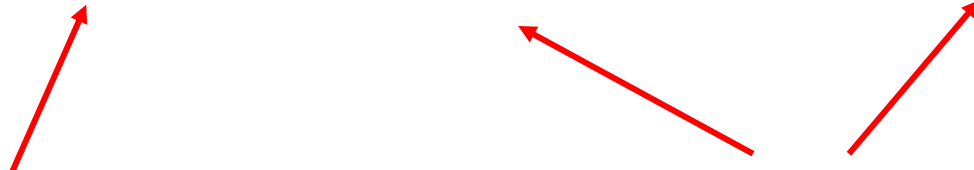
Up to scale

Conversion

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

Homogeneous Coordinates

$$T \begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} Rx + p \\ 1 \end{bmatrix}$$


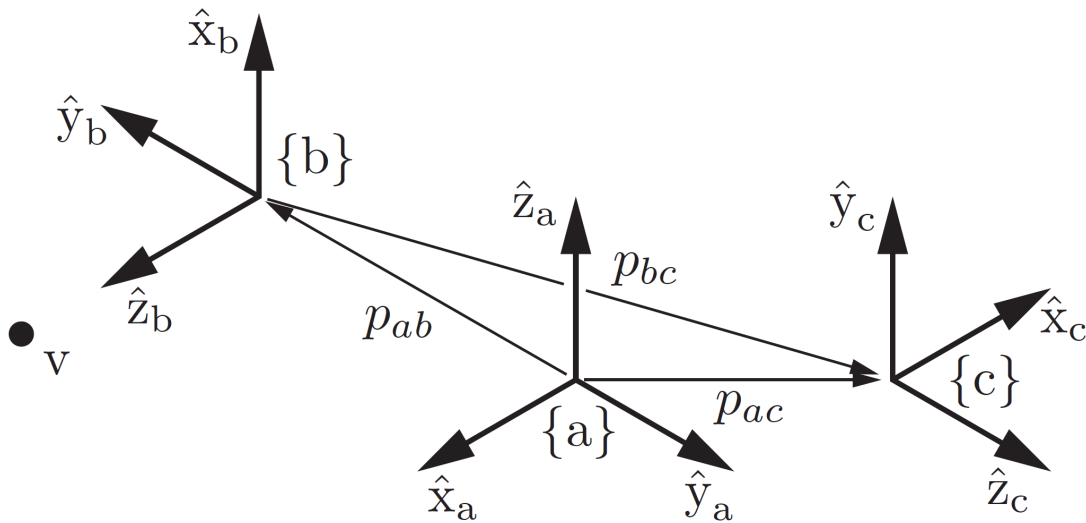
Homogeneous transformation

Homogeneous coordinates

Uses of Transformation Matrices

- Represent the configuration of a rigid-body
- Change the reference frame
- Displace a vector or a frame

Representing a Configuration



$$R_{sa} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_{sb} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad R_{sc} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$p_{sa} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad p_{sb} = \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix}, \quad p_{sc} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$T_{sa} = (R_{sa}, p_{sa}) \quad T_{sb} = (R_{sb}, p_{sb})$$

$$T_{sc} = (R_{sc}, p_{sc})$$

$$T_{bc} = (R_{bc}, p_{bc}) \quad R_{bc} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{bmatrix}$$

$$p_{bc} = \begin{bmatrix} 0 \\ -3 \\ -1 \end{bmatrix}$$

$$T_{de} = T_{ed}^{-1}$$

Changing the Reference Frame

$$T_{ab}T_{bc} = T_{a\cancel{b}}T_{\cancel{b}c} = T_{ac}$$

$$T_{ab}v_b = T_{a\cancel{b}}v_{\cancel{b}} = v_a$$

Displacing a Vector or a Frame

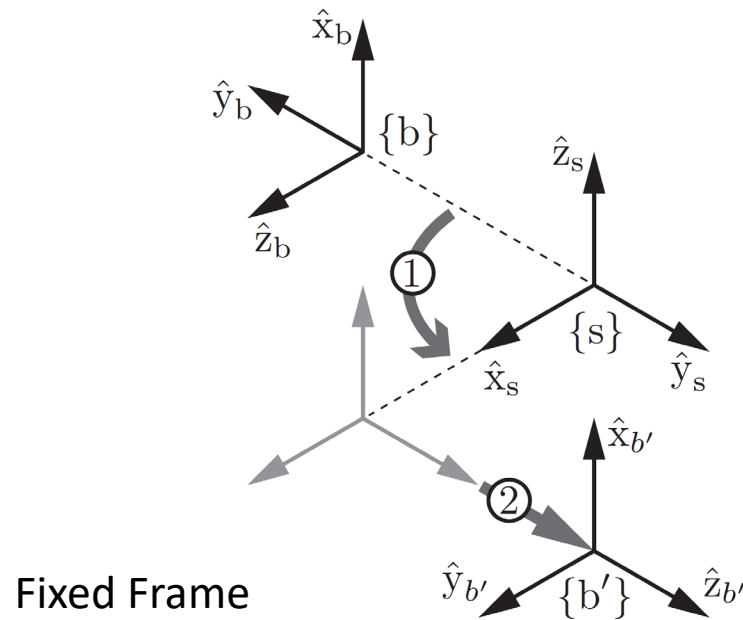
- Rotating and then translating $(R, p) = (\text{Rot}(\hat{\omega}, \theta), p)$
- Transformation matrices

$$\text{Rot}(\hat{\omega}, \theta) = \begin{bmatrix} R & 0 \\ 0 & 1 \end{bmatrix} \quad \text{Trans}(p) = \begin{bmatrix} 1 & 0 & 0 & p_x \\ 0 & 1 & 0 & p_y \\ 0 & 0 & 1 & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Displacing a Vector or a Frame in Fixed Frame

$$T_{sb'} = TT_{sb} = \text{Trans}(p) \text{Rot}(\hat{\omega}, \theta) T_{sb} \quad (\text{fixed frame})$$

$$= \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_{sb} & p_{sb} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} RR_{sb} & Rp_{sb} + p \\ 0 & 1 \end{bmatrix}$$

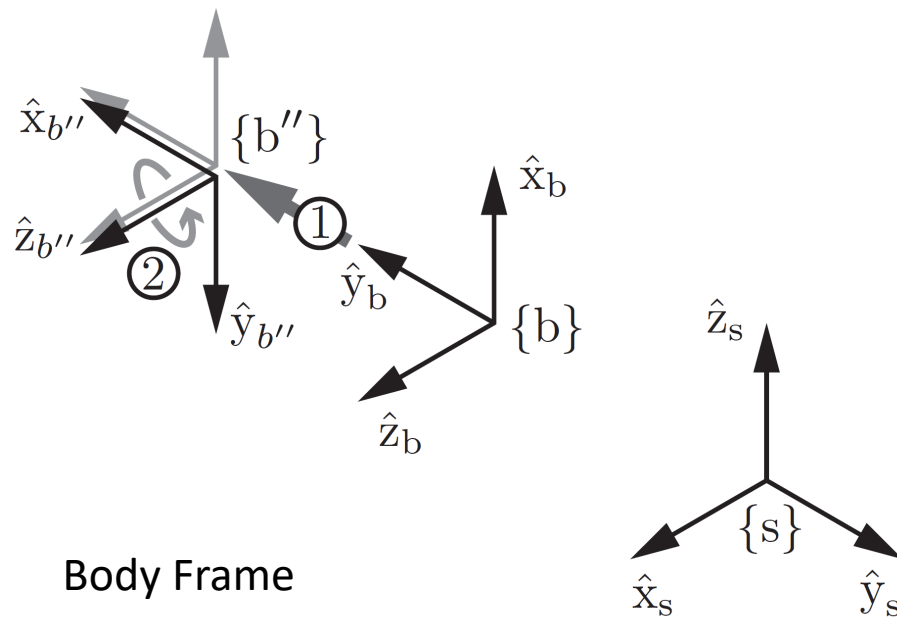


$$\hat{\omega} = (0, 0, 1) \quad \theta = 90^\circ \quad p = (0, 2, 0)$$

Displacing a Vector or a Frame in Body Frame

$$T_{sb''} = T_{sb}T = T_{sb} \text{Trans}(p) \text{Rot}(\hat{\omega}, \theta) \quad (\text{body frame})$$

$$= \begin{bmatrix} R_{sb} & p_{sb} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R_{sb}R & R_{sb}p + p_{sb} \\ 0 & 1 \end{bmatrix}$$



$$\hat{\omega} = (0, 0, 1) \quad \theta = 90^\circ \quad p = (0, 2, 0)$$

Summary

- Reference frames
- Rigid-body in 3D
 - Rotation matrices
- Homogenous transformation matrices
- Uses of Transformation Matrices

Further Reading

- Chapter 3 in Kevin M. Lynch and Frank C. Park. Modern Robotics: Mechanics, Planning, and Control. 1st Edition, 2017
- Quaternion and Rotations, Yan-Bin Jia,
<https://graphics.stanford.edu/courses/cs348a-17-winter/Papers/quaternion.pdf>
- [On the Continuity of Rotation Representations in Neural Networks.](#)
Zhou et al., CVPR, 2019.